Baloyi Thabang Bongani Junior

Quantitative Modelling | *Mathematical Statistics* | *AI & Data Engineering*

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PROFESSIONAL SUMMARY

Quantitative and AI engineering professional with an MSc in Mathematical Statistics completed with distinction, supported by academic training in Statistics and Data Science, Actuarial Science, and quantitative finance. Experience spans actuarial pricing and risk assessment, statistical modelling, model validation, data engineering, production AI systems, and full-stack software development.

Skilled in Python, R, SQL, C#, VBA, TensorFlow, scikit-learn, time series analysis, machine learning, optimisation, REST APIs, testing, CI/CD, and cloud analytics. Built decision-support systems combining SAP-derived planning data, demand and capacity modelling, mixed-integer optimisation, and AI-assisted workflows.

Combines rigorous quantitative analysis with end-to-end engineering and clear business communication to investigate model behaviour, validate outputs, automate analytical workflows, and translate complex results into practical decisions.

EDUCATION

University of the Free State — **Bloemfontein, ZA**
Master of Science, Mathematical Statistics — *Jan 2026 – Oct 2026*
- GPA: 80% (WITH DISTINCTION) | UFS Academic Record

University of Cape Town — **Cape Town, ZA**
Bachelor of Science Honours, Statistics and Data Science, minor in Actuarial Science — *Dec 2025*
- GPA: 74% | UCT Transcript Honours | Proof of Qualification

University of Cape Town — **Cape Town, ZA**
Bachelor of Commerce, Management Studies — *Dec 2022*
- GPA: 70% | UCT Transcript Undergraduate | UCT Certificate

University of the Free State — **Bloemfontein, ZA**
Bachelor of Science, Actuarial Sciences (Incomplete) — *2016 – 2017*
- Not completed | GPA: 80% | UFS Academic Record

TECHNICAL SKILLS

Programming & Data Science: Python, R, SQL, MATLAB, STATA, Java, C#, VBA, LaTeX, TensorFlow, Keras, scikit-learn, NumPy, pandas, matplotlib, Spark, PySpark, Databricks
Statistical & Analytical: Statistical Modelling, Regression, Classification, Time Series Analysis, Multivariate Analysis, Bayesian Analysis, Machine Learning, Neural Networks, Random Forests, Portfolio Theory, Hypothesis Testing, Monte Carlo Simulation, Mixed-Integer Programming, Linear Programming, Critical Path Analysis, Supply Chain Optimisation, Root Cause Analysis
Data, APIs & BI: PostgreSQL, Amazon Redshift, Snowflake, Delta Lake, dbt, pgvector, Azure Table Storage, Azure Blob Storage, Azure Data Lake Storage Gen2, REST APIs, Flask, Power BI, Tableau, Microsoft Excel, Data Storytelling, KPI Reporting
Cloud, DevOps & Tools: AWS (EC2, S3, Lambda, Glue, Redshift), Azure (VMs, App Service, Container Apps, OpenAI, Blob Storage, Data Factory, Synapse Analytics, Fabric), Docker, Git, GitHub Actions, Bitbucket Pipelines, Azure DevOps, SonarQube, Jira, Jupyter Notebook, VS Code, Postman

WORK EXPERIENCE

Senior Full Stack AI Engineer
4S4, Cape Town, ZA (Full Time, Hybrid) — *Jun 2026 – Present*
- **Clients:** Rheinmetall Denel Munition, Old Mutual Bank, Ooba, Deloitte
- Led development of EAPP, an enterprise demand and production-planning platform that integrates SAP-derived demand, stock and open supply, multi-level BOM explosion, finite work-centre capacity, lead-time calculation, inventory optimisation, and cross-plant scheduling

- Built explainable optimisation workflows using priority-based allocation, critical-path analysis, Monte Carlo lead-time simulation, linear programming, and lexicographic mixed-integer programming to identify capacity constraints and recommend feasible production, overtime, transfer, and delivery plans
- Implemented Tiffany, an embedded voice and text AI planning co-pilot integrating OpenAI speech and realtime services with Anthropic Claude, registered planning commands, structured outputs, audit trails, and human confirmation for governed actions
- Develop and maintain data integration pipelines, including SAP S/4HANA 2023-to-warehouse extraction, ETL workflows, and semantic layers
- Build and orchestrate agentic and automation workflows using n8n, multi-agent systems, and RAG pipelines
- Collaborate with client stakeholders to translate business requirements into technical solutions and own features across architecture, implementation, testing, deployment, and support
- Support DevOps practices including CI/CD, Azure cloud deployment, and version-control workflows

Artificial Intelligence Engineer
Full Stack, Cape Town, ZA (Full Time) *Mar 2023 – May 2026*

- **Clients:** Virgin Active : Virgin Active UK and Virgin Active SA, Expandly
- Design and build AI-driven, cloud-native applications on AWS and Azure, integrating LLM services, data workflows, and Keras-based models into production systems
- Develop backend systems, REST APIs, and data services for RAG pipelines, embeddings, semantic search, profile management, recommendation workflows, and analytics-ready outputs
- Build data and ML workflows using Python, SQL, Spark, PostgreSQL, pgvector, Azure Table Storage, Azure OpenAI, Databricks, PySpark, Delta Lake concepts, and OLLAMA-based embedding services
- Support cloud data engineering patterns including data pipeline development, data modelling, batch processing, lakehouse-style organisation, and medallion architecture principles
- Implement CI/CD-supported testing, build, code quality, and deployment workflows using Bitbucket Pipelines, GitHub Actions, Azure DevOps, pytest, SonarQube, and endpoint smoke tests
- Integrate AI and ML models into Flask services, production APIs, and cloud-hosted backend applications with monitoring, validation checks, and data quality controls
- Build AI safety and compliance components, including guardrails, risk-pattern detection, medical caution logic, emergency response routing, and auditable decision workflows
- Collaborate with testers, engineers, product stakeholders, and business users on architecture decisions, QA validation, cloud design, root cause analysis, and AI/data platform strategy

AI Machine Learning Engineer Intern
Full Stack, Cape Town, ZA *Jan 2023 – Mar 2023*

- Performed exploratory data analysis on e-commerce and member engagement datasets using pandas, NumPy, matplotlib, and SQL to identify patterns for model development and data quality improvement
- Built and evaluated classification and regression models, including Logistic Regression, Random Forests, and XGBoost, using scikit-learn
- Developed preprocessing and feature engineering pipelines for missing-value handling, categorical encoding, feature scaling, data validation, and model-ready data preparation
- Assisted with Azure OpenAI integrations in backend applications, including prompt engineering, response parsing, prototype conversational AI workflows, and cloud-hosted data persistence
- Created data visualisations and analytical reports using matplotlib and Power BI to communicate model performance, data patterns, and business insights
- Supported Flask REST API deployment on Azure App Service, endpoint testing with Postman, and basic CI/CD validation
- Conducted end-to-end automation testing using WebdriverIO, covering authentication, dashboard interactions, and data validation across web applications

Group Risk Intern
Discovery Limited, Cape Town, ZA *Jan 2021 – Dec 2022*

- Supported actuarial pricing and risk assessment activities within the Group Risk reinsurance environment
- Analysed insurance and reinsurance exposures, claims, premiums, and risk-related data to support pricing adequacy and actuarial decision-making

- Assisted with the assessment of reinsurance arrangements, including retained and ceded risk, expected losses, and underlying risk exposure
- Supported pricing calculations using actuarial and statistical methods and evaluated loss ratios, expected claims, and other actuarial measures
- Performed sensitivity and scenario analysis to assess the impact of changes in key pricing and risk assumptions
- Contributed to the validation and review of actuarial model outputs and assumptions, helping identify inconsistencies and support robust quantitative results
- Prepared analytical summaries and reports communicating pricing and risk findings to relevant stakeholders
- Gained practical exposure to reinsurance, actuarial modelling, insurance risk management, and financial services governance

PROJECTS

- **EAPP / Tiffany Strategic Planning Co-Pilot** – Built a full-stack planning application integrating SAP-derived demand, sales-order traceability, stock and supply netting, recursive BOM explosion, finite-capacity planning, backward scheduling, and lead-time estimation. Developed MILP/MIP, critical-path, and Monte Carlo optimisation workflows for production, overtime, inventory, transfers, and delivery plans. Embedded a voice and text AI co-pilot with governed, audited commands. Tech stack: React, Python, SQL, Azure Container Apps, OpenAI Realtime and speech APIs, Anthropic Claude, HiGHS. Live Demo
- **Virgin Active AI Wellness Coach** – Built an Azure-hosted conversational fitness coach delivering personalised workouts, classes, adaptive coaching, progressive profiles, and structured goals. Implemented recommendation data flows, medical-clearance guardrails, real-time streaming, and persistent sessions. Tech stack: Python, Flask, Azure OpenAI, Azure App Service, Azure Table and Blob Storage, PostgreSQL, pgvector, REST APIs. Live Demo
- **ExpandlyAI** – Developed an enterprise e-commerce automation platform integrating Amazon SP-API with C#/.NET 10 services and an Angular Nx frontend. Implemented an Azure OpenAI proxy, cost-calculation services, Azure Table Storage persistence, WebdriverIO end-to-end tests, and Azure DevOps CI/CD.

ACTUARIAL EXEMPTIONS (ASSA)

A112 – Business Economics
A113 – Business Finance

CERTIFICATIONS

- CFA Level I (In Progress)

HONOURS & AWARDS

- Dean's List, University of the Free State
- Best Aggregate Average Actuarial Science, University of the Free State
- Best Aggregate Average Actuarial Science, Moshal Scholarship
- GoldenKey International Society
- Top 3 Learner at Nghonyama High School, Grade 1–12
- Moshal Scholarship
- Jurgen Klauss Bathe Leadership Scholarship

VOLUNTEERING

- **Tutor, Grade 8 and 9** – Adopt-a-school Foundation, 2021
- **UCT Mathematics Competition** – 2019 – 2023

REFERENCES

Additional references available upon request.